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GITHUB -

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A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. For a given channel, it can be assumed that a transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0. If a signal is sent, determine the probability that:

R = Received, T = Transmitted

```
binary_comm <- data.frame(correct = c(0.95, 0.75), error = c(0.05, 0.25),  
                           sent = c(0.70, 0.30), row.names = 0:1)  
binary_comm
```

```
##   correct error sent  
## 0    0.95  0.05  0.7  
## 1    0.75  0.25  0.3
```

(a) A 1 was received:

$$P(R = 1) = P(R = 1|T = 0) \cdot P(T = 0) + P(R = 1|T = 1) \cdot P(T = 1)$$

```
prob_get1 <- round((0.05 * 0.7 + .75 * 0.3), 4)  
cat(prob_get1 * 100, "%", sep = "")
```

```
## 26%
```

(b) A 1 was transmitted given then a 1 was received:

$$P(T = 1|R = 1) = \frac{P(T=1) \cdot P(R=1|T=1)}{P(R=1)}$$

```
prob_send1_get1 <- round(((0.3 * 0.75) / prob_get1), 4)  
cat(prob_send1_get1 * 100, "%", sep = "")
```

```
## 86.54%
```

7. There are three employees working at an IT company: Jane, Amy, and Ava, doing 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and just 1% of Ava's work is in error. What is the overall percentage of error? If a program is found with an error, who is the most likely person to have written it?

```
## Given probabilities
P_Jane <- 0.10
P_Amy <- 0.30
P_Ava <- 0.60

## Probability that each employee's work contains an error
P_Error_Jane <- 0.08
P_Error_Amy <- 0.05
P_Error_Ava <- 0.01

## Calculate the overall probability of an error
P_Error <- (P_Jane * P_Error_Jane) +
  (P_Amy * P_Error_Amy) +
  (P_Ava * P_Error_Ava)

print(P_Error * 100) # Overall percentage of errors
```

```
## [1] 2.9
```

```
## Calculate the probability that each employee wrote a program with an error
P_Jane_Error <- (P_Jane * P_Error_Jane) / P_Error
P_Amy_Error <- (P_Amy * P_Error_Amy) / P_Error
P_Ava_Error <- (P_Ava * P_Error_Ava) / P_Error

print(P_Jane_Error) # Probability the error was made by Jane
```

```
## [1] 0.2758621
```

```
print(P_Amy_Error) # Probability the error was made by Amy
```

```
## [1] 0.5172414
```

```
print(P_Ava_Error) # Probability the error was made by Ava
```

```
## [1] 0.2068966
```